

ROSKILDE AP, Denmark

WMO: 061700

Lat: 55.587N

Lon: 12.136E

Elev: 146

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	19.5	11.5	10.0	18.9	15.0	11.8	22.0	33.7	41.6	30.4	40.9	6.0	350	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	78.7	65.0	75.5	64.4	73.0	63.2	68.0	74.5	66.2	72.7	64.5	70.8	10.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	95.8	71.1	64.0	89.6	69.4	62.2	84.2	67.6	32.4	74.9	30.9	72.9	29.7	70.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	24.2	21.6	DB	8.0	84.2	6.6	3.4	3.2	86.7	-0.6	88.6	-4.3	90.5	-9.1	93.0	(4)
(5)			WB	7.6	70.9	6.6	1.8	2.9	72.2	-1.0	73.3	-4.6	74.3	-9.4	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	33.1	33.6	37.2	44.7	52.0	57.7	62.6	62.5	56.4	48.6	41.1	35.6		
	DBStd	12.09	6.97	6.73	6.24	5.90	5.57	4.74	4.75	4.59	4.56	5.46	5.78	7.17		
	HDD50	2421	525	460	398	179	42	2	0	0	6	92	271	446		
	HDD65	6579	990	880	862	609	403	223	107	107	260	509	718	911		
	CDD50	1385	0	0	1	20	105	234	390	387	197	48	3	0		
	CDD65	68	0	0	0	0	1	5	32	29	1	0	0	0		
	CDH74	444	0	0	0	1	7	49	230	152	5	0	0	0		
	CDH80	53	0	0	0	0	0	8	25	20	0	0	0	0		
(14) Wind	WSAvg	10.6	12.2	12.3	11.8	10.7	9.7	9.8	9.0	9.4	10.0	10.6	10.9	11.3		
(15) Precipitation	PrecAvg	24.8	1.9	1.5	1.4	1.3	1.8	2.4	2.4	2.8	2.5	2.6	2.2	2.2		
	PrecMax	31.8	3.6	3.4	3.1	3.0	3.5	5.5	6.1	7.4	6.0	5.0	4.8	4.1		
	PrecMin	16.7	0.1	0.2	0.3	0.2	0.5	0.0	0.2	0.6	0.6	1.0	0.5	1.0		
	PrecStd	3.7	0.9	0.6	0.7	0.6	0.7	1.3	1.4	1.7	1.5	1.1	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	50.2	57.1	67.8	74.6	78.9	82.8	82.6	73.8	63.6	55.6	51.5		
		MCWB	47.3	47.1	48.8	56.4	62.5	65.3	67.1	66.8	63.0	58.4	53.3	49.8		
	2%	DB	46.3	47.4	52.0	62.3	70.0	74.4	79.2	77.9	69.8	60.4	53.2	49.2		
		MCWB	45.1	45.5	46.1	53.0	59.0	63.3	65.1	65.4	61.6	56.5	51.3	47.7		
	5%	DB	44.4	45.0	48.5	58.4	66.3	70.6	76.7	74.9	66.5	58.1	51.1	46.7		
		MCWB	43.1	43.2	44.7	50.6	56.8	61.4	64.6	64.6	60.0	55.1	49.3	45.5		
	10%	DB	42.5	42.9	46.2	55.0	63.0	67.7	73.3	71.7	64.3	56.2	48.8	44.8		
		MCWB	41.2	41.3	43.4	48.5	55.1	59.2	63.7	63.2	58.5	53.5	47.4	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	47.9	50.5	57.7	63.8	68.6	70.4	70.1	65.4	59.5	54.0	50.1		
		MCDB	48.6	48.9	54.8	65.3	71.9	76.3	77.8	76.5	71.1	62.3	55.0	50.8		
	2%	WB	44.9	45.6	47.3	54.1	60.8	64.8	68.3	67.8	62.9	57.3	51.6	47.8		
		MCDB	45.7	46.9	50.4	60.5	67.1	71.7	75.4	73.6	67.6	59.5	52.8	48.8		
	5%	WB	43.0	43.4	45.3	51.5	58.2	62.6	66.5	66.0	61.2	55.6	49.5	45.5		
		MCDB	43.8	44.7	47.7	56.6	64.4	68.6	73.3	72.1	65.4	57.7	50.5	46.4		
	10%	WB	41.0	41.2	43.2	49.2	55.9	60.4	64.8	64.4	59.5	54.2	47.6	43.6		
		MCDB	41.9	42.6	45.6	53.7	61.7	66.0	71.2	70.3	63.2	56.2	48.5	44.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	11.6	14.9	16.4	16.1	16.7	15.6	13.3	10.6	8.1	7.8		
		MCDBR	7.8	9.6	15.4	19.8	21.4	20.2	21.7	19.3	16.2	11.7	8.9	8.0		
	5% WB	MCWBR	7.5	7.8	10.8	11.9	12.6	11.4	10.5	9.7	9.7	8.8	7.7	7.8		
		MCWBR	7.9	9.3	14.2	18.1	18.7	18.5	18.4	16.8	14.4	10.8	8.4	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.335	0.359	0.393	0.388	0.382	0.404	0.404	0.374	0.355	0.335	0.303		
		taud	2.290	2.312	2.304	2.266	2.340	2.383	2.341	2.349	2.415	2.405	2.331	2.262		
	Ebn at Noon		189	231	254	262	271	274	264	256	248	222	179	162		
		Edh at Noon	17	24	31	37	36	36	36	34	28	22	17	14		
(44) All-Sky Solar Radiation	RadAvg		143	334	779	1324	1702	1818	1688	1354	959	473	182	98		
	RadStd		26	53	82	160	166	138	196	97	94	66	30	10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-252	N/A	N/A
(47) Regional (66 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-290	N/A	N/A

Nomenclature: See separate page